

In-process QC Procedure For Onboard Kavach Installation

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In-process QC Procedure for Onboard Kavach Installation	In-process QC Procedure for Onboard Kavach Installation	22	25	-

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Abstract

This document describes the standard procedure of In-process QC Procedure for Onboard Kavach Installation

DOCUMENT CONTROL SHEET

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3	G Prabhakar	HBL	Sr Manager QA	Prepared

CHANGE HISTORY

#	Name of the Document	Date	Reason for changes	Version
1	In-process QC Checklist for Installation of KAVACH	06.06.2023	Initial Version	1.0
2	In-process QC Checklist for Installation of KAVACH	10.07.2023	Software files Checksum Verification Added.	1.1
3	In-process QC Checklist for Installation of KAVACH	30.12.2023	RFID Assembly details updated	1.2
4	In-process QC Checklist for Installation of KAVACH	01.11.2024	Added CCB Installation details	1.3
5	In-process QC Checklist for Installation of KAVACH	09.09.2025	Procedure updated for WAG-9 & WAP-7 Locos as per the CLW Installations	1.4

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1. Purpose

The purpose of this manual is to specify Onboard KAVACH equipment Quality Inspection procedure to inspect quality issues in Locomotives.

2. Scope

Scope of this document is to lay down the quality inspection procedure of onboard KAVACH and its sub-component units during installation and commissioning. Quality incharge shall be responsible for maintaining this Procedure.

3. Quality Check Points:

3.1 Onboard KAVACH:

- 3.1.1 Confirm that the location of the Loco Kavach unit in the loco allows access for easy replacement of PCBs.
- 3.1.2 Check the Loco Kavach unit stand welding without any gaps with respect to the loco surface.
- 3.1.3 Ensure that there is no loose joint between stand to surface during welding.
- 3.1.4 Ensure that finishing of the welding to be smooth and there is no sharp edges.



Figure 1: Loco KAVACH Stand Mounting

- 3.1.5 The welded loco stand shall be made of 5mm thick steel, to withstand Loco vibrations. The structure shall be stable and standing on four legs. Cantilever structures with two legs shall not be used.
- 3.1.6 The stand shall be treated with red oxide coating, before being painted with RAL7032 (Smoke Grey/Pebble Grey color) paint, to avoid corrosion.
- 3.1.7 Ensure that Loco stand fixing holes(M8) are exactly matching with Loco channel fixing holes(M8).
- 3.1.8 The M8 bolts shall be tightened with 25N-m torque, using a torque spanner. The bolt heads shall be marked with green paint, after the QC inspector checks for proper torque application.
- 3.1.9 Check the cable connections are made as per the labeling without overlaps and wrong connections.
- 3.1.10 Ensure that all the external cable circular connectors are fully locked properly with Loco Kavach receptacles.

Improper Cable Routing



Proper Cable Routing



Figure 2: Loco Kavach unit cables routing

3.2 EMI Filter Box:

- 3.2.1 Ensure that fixing of EMI Filter Box to loco stand with (M5x16mm) screws and nuts with torque wrench and maintained torque 6N-m for proper tightness of fixing.
- 3.2.2 Check the cable routing and ensure that there are no sharp bends and stress on cables.
- 3.2.3 Ensure all the cables routed through PG glands and also check all the cables are tied with cable ties.

Proper Cable Routing**Improper routing of EMI filter cable**

Figure 3: EMI Filter Unit Mounting

3.3 RIB and Cab Input Box:

- 3.3.1 Ensure that enough space available in-between cab input box and RIB unit for easy access of cables.
- 3.3.2 Check the RIB and CAB Input box stand welding should be proper without any gap on welding.
- 3.3.3 The welded loco stand shall be made of 5mm thick steel, to withstand Loco vibrations.
- 3.3.4 Welding portion to be clean and smooth without any cracks and joint breaks.
- 3.3.5 The stand shall be treated with red oxide coating, before being painted with RAL7032 (Smoke Grey/Pebble Grey color) paint, to avoid corrosion.
- 3.3.6 The M5X16mm bolts shall be tightened with 6N-m torque, using a torque spanner. The bolt heads shall be marked with green paint, after the QC inspector checks for proper torque application.



Figure 4: RIB & CAB I/P BOX Mounting

- 3.3.7 Cables to be routed without any sharp bends, overlaps and tied properly with cable ties.
- 3.3.8 Ensure that cables are connected with respect to labels and ensure circular connectors are fully locked with enclosure receptacles.



Figure 5: RIB & CAB I/P Cable Routing

3.4 DMI/LP-OCIP Mounting:

- 3.4.1 Check the DMI mounted place is good enough at driver desk, which can be operated easily by Loco pilot.
- 3.4.2 Ensure that DMI mounting stand is properly welded without any joint gaps. Ensure that mounting stand can be withstand to loco vibrations.
- 3.4.3 The M5X16mm screws shall be tightened with 6N-m torque, using a torque spanner. The screws heads shall be marked with green paint,

after the QC inspector checks for proper torque application.

3.4.4 Ensure that projection at bottom side of the stand to be easy to access DMI cable.

3.4.5 The stand shall be treated with red oxide coating, before being painted with RAL7032 (Smoke Grey/Pebble Grey color) paint, to avoid corrosion.

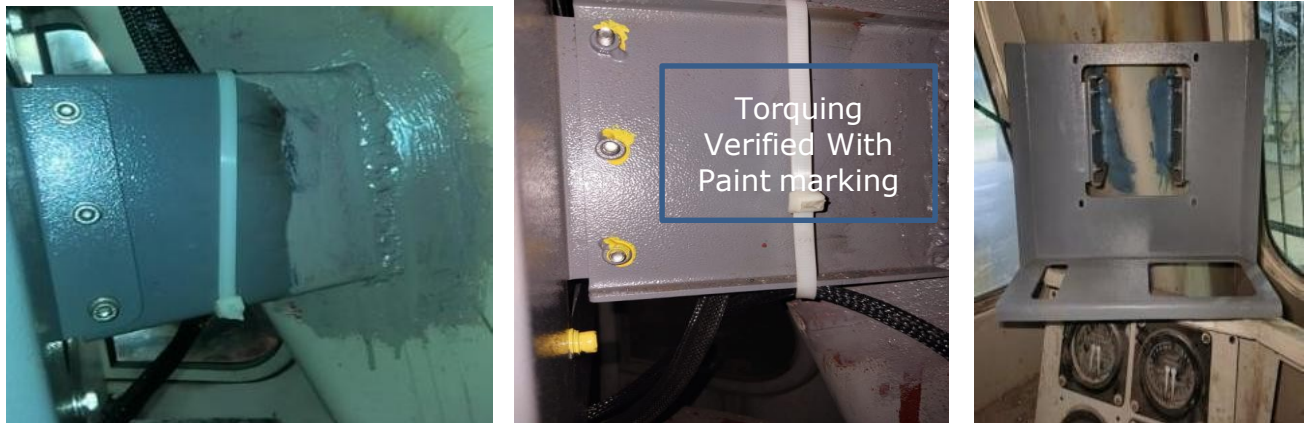


Figure 6: DMI/LP-OCIP Mounting stand

3.4.6 Ensure that cable routing done properly without over stress and sharp bends.

3.4.7 Ensure that the cable booting is not damaged/peeled out while or after routing the cable.

3.4.8 DMI-1 cable shall be connected to MC1 and DMI-2 cable shall be connected to MC3 at Loco Kavach unit.

3.4.9 Ensure that the circular connectors are fully locked with DMI unit enclosure receptacles.



Figure 7: DMI/LP-OCIP Cable Routing

3.5 RFID PS Box:

- 3.5.1 The M5X16mm screws shall be tightened with 6N-m torque, using a torque spanner to Loco Kavach stand. The screws heads shall be marked with green paint, after the QC inspector checks for proper torque application.
- 3.5.2 Ensure that cable connections are given at the corresponding location with respect to labels provided at Loco Kavach unit.
- 3.5.3 Ensure that cables are routed properly without hanging and tied properly with cable ties.
- 3.5.4 Ensure the circular connectors are properly locked with RFID box unit receptacles.



Figure 8: RFID PS Box Fixing and Cable Routing

3.6 Radio & GPS-GSM Antenna's:

- 3.6.1 Check the Radio antenna's base plate welding done properly without any joint gaps and cracks.



Figure 9: Radio(RF) Antenna Fixing

3.6.2 Ensure that welding done good enough, such that antenna's can be withstand loco vibrations, while running with higher speeds.



Figure 10: GPS&GSM Antenna Fixing

- 3.6.3 Ensure that antenna's are mounted within stipulated height to avoid OHE line contact.
- 3.6.4 Ensure that the LMR400(TX-1,TX-2,RX-1&RX-2)&LMR200(GSM1, GPS1, GSM2& GPS2)cables are connected to their respective locations.
- 3.6.5 Ensure that the cables are routed properly through the 2" steel reinforced pipe/hose to avoid environmental damages and pipe/hose tied to supporting welded rods using cable ties.
- 3.6.6 Ensure that cables are not exposed outside after routing through reinforced pipe/hose. Any locally procured conduit shall be of **KONTACT**(or Equivalent) make and steel wire reinforced pipe model.
- 3.6.7 Ensure that the all welded portions shall be treated with red oxide coating, before being painted with RAL7032 (Smoke Grey/Pebble Grey color) paint, to avoid corrosion

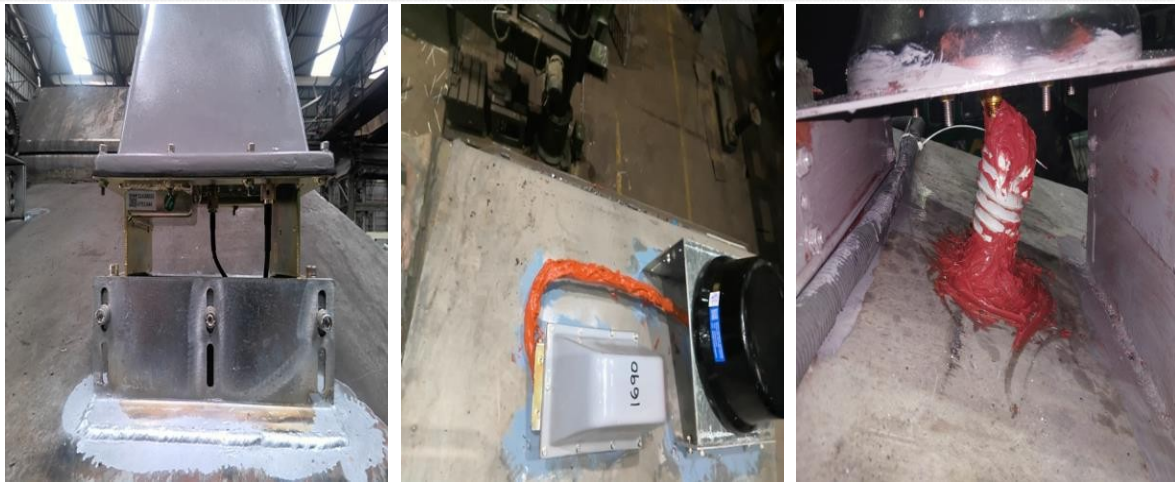


Figure 11: Radio, GSM& GPS antenna cable routing

3.7 Pneumatic fittings and EP Valve, Cocks fixing :

- 3.7.1 Confirm that all pipes and fittings used in the assembly are from the approved BOM, and taken from the I&C kit supplied from the factory. Any locally procured items must be checked for compliance to the approved BOM.
- 3.7.2 Confirm that copper pipes are bent using appropriate bending tool, and that there no kinks or sharp bends in the pipe.
- 3.7.3 Ensure that copper tube length is measured with respect the connectivity from loco pneumatics to EP Valve, BP cock, Horn cock and valve arrangements.

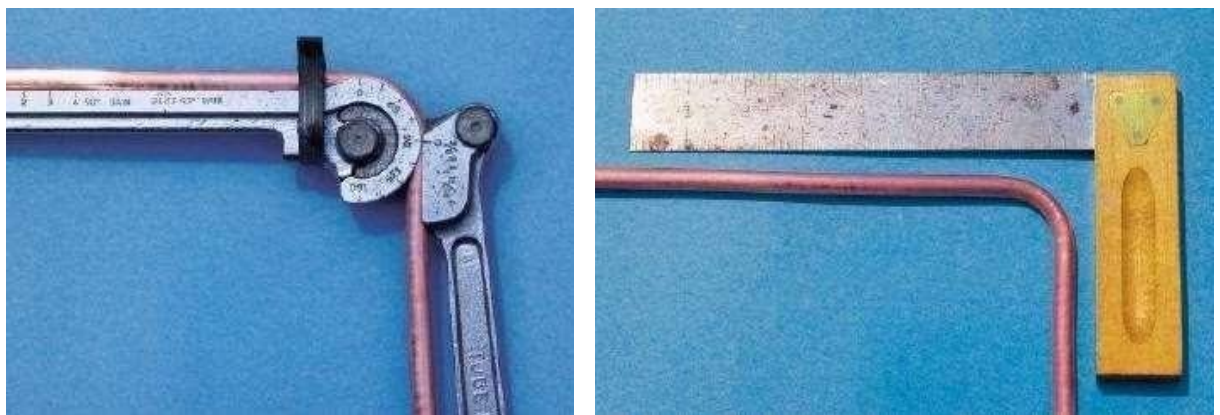


Figure 12: Copper pipe bending process with bend tool

- 3.7.4 Ensure that copper pipe connections made properly with approved make(Ex.Fluid Control)ferrules and TEE-joints used.
- 3.7.5 All threaded connections must be sealed with Teflon paste loctite 567 (NOT A TEFLON TAPE), which is supplied in the I&C kit.



Figure 13: Pneumatic's assembly and ferrules fixing

3.7.6 Ensure that there are no loose connections in the pneumatic lines, by checking with soap solution. No air bubbles shall be seen when soap solution is applied at the joints. After the test, the soap solution shall be wiped clean.

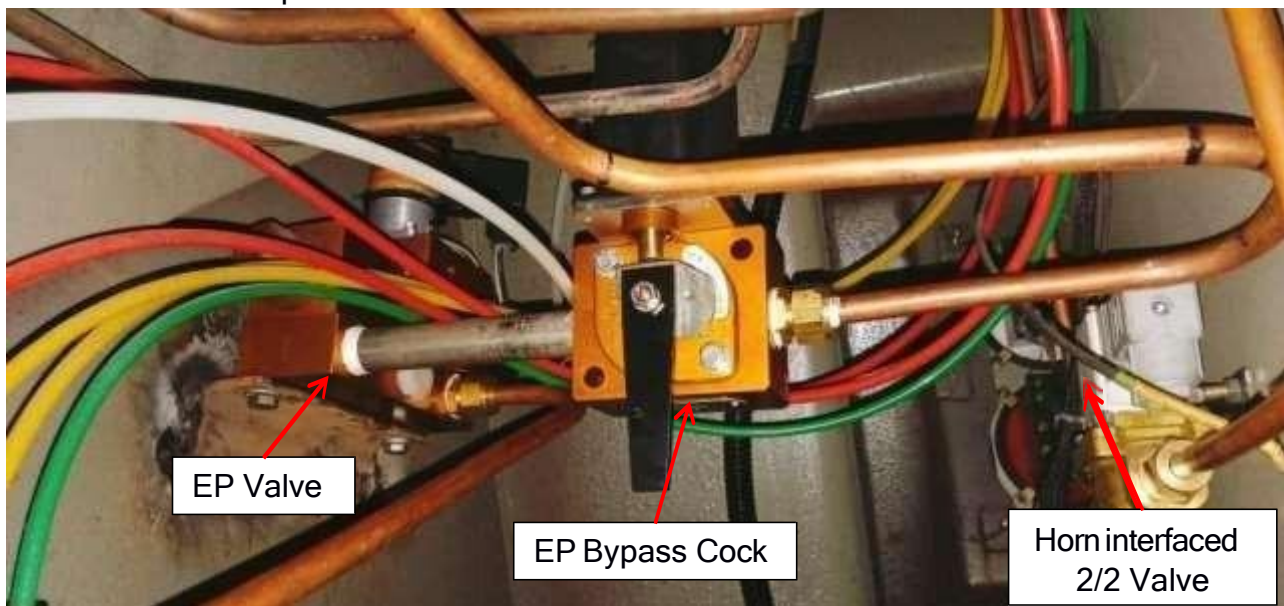


Figure 14: EP Valve, EP Cock and Horn valve assembly

3.8 Installation of Light Engine (LE) Unit:

- 3.8.1 D2 relay valve and existing manifold unit shall be removed from E-70 brake panel.
- 3.8.2 LE unit, supplied by factory, shall be installed in place of manifold unit, using the removed hardware. The removed manifold shall be handed over to the Loco Shed officials.
- 3.8.3 D2 relay valve shall be installed on LE unit.

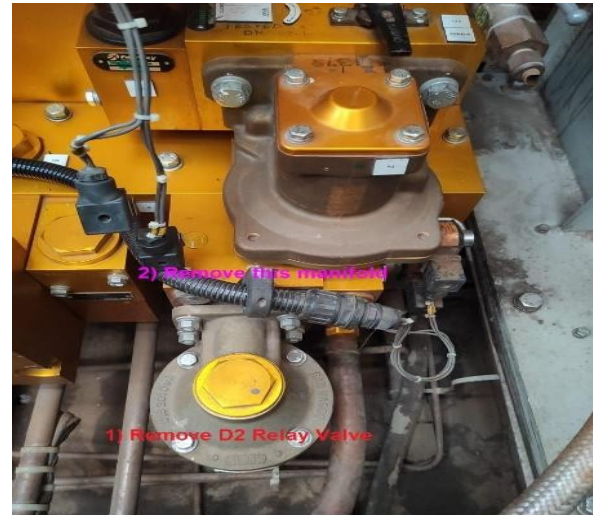


Figure 15: Light Engine (LE) Unit

3.9 MR, BP, BC1 & BC2 Pressure sensors fixing at CAB-I:

- 3.9.1 Ensure the MR sensor to be installed on MR pressure gauge copper pipe with 1/2" T-Joint provided with I&C kit, connected with "RED" color pipe. Ensure the sensor range shall be **0-16 Bar**.
- 3.9.2 Ensure the BP sensor to be installed on BP pressure gauge copper pipe with 3/8" T-Joint, connected with "GREEN" color pipe. Ensure the sensor range shall be **0-7 Bar**.
- 3.9.3 The BC-1 & BC-2 sensors shall be installed to BC pressure gauges, connected with "YELLOW" color two individual pipes for BC-1 & BC-2. Ensure the two BC sensors range shall be **0-7 Bar**.
- 3.9.4 Ensure these sensors wiring connection been given as per the labels specified in MC26 connector of RIB external harness.
- 3.9.5 Following image illustrates the Pressure sensors connections.

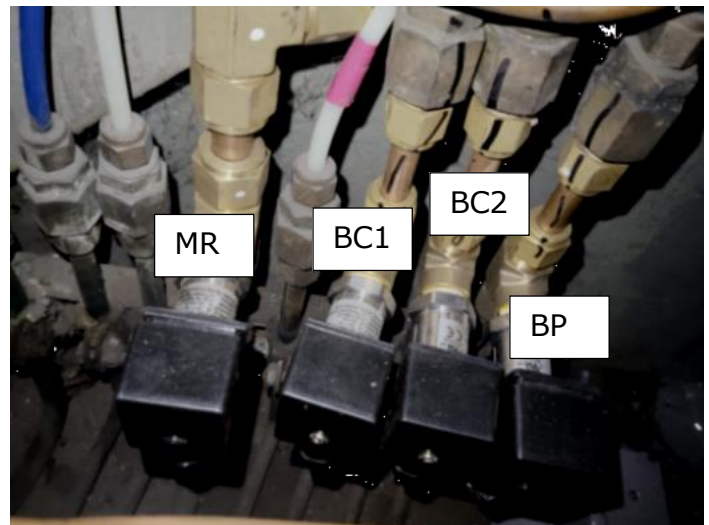


Figure 16: MR, BP, BC1 & BC2 sensor installation

3.10 IRU(Faively) units Fixing:

- 3.10.1 Check and Ensure that sufficient place available for IRU fixing at bottom of the driver desk of both the CAB's. (NOTE: Ensure that ISO COCK SHOULD BE CLOSED CONDITION WHILE INSTALLATION)
- 3.10.2 Ensure that supporting angles for IRU units are welded properly to withstand loco vibrations.
- 3.10.3 Ensure that welded portion should be neat and clean. Check that there is no welding gaps and cracks.



Figure 17: Interface Relay Unit (IRU) assembly

- 3.10.4 Ensure that the IRU units fixing holes are matching with supporting angels.
- 3.10.5 Ensure that the welded portion shall be painted with RAL7032 (Smoke Grey/Pebble Grey color) paint, to avoid corrosion.
- 3.10.6 Check the cables are connected to their respective position.
- 3.10.7 Ensure that cables are routed properly without hanging on ground and tied properly with cable ties.

3.11 PG's and Speed meter boxes fixing:

- 3.11.1 Ensure that PG1 and PG2 are installed on allotted axles(1/2/3/4/5/6) of locomotives on left and right side.
- 3.11.2 Ensue the PG1 is connected left side Wheel No.5 and PG2 on right side wheel no.2 , while sitting from the LP desk at CAB-A side.

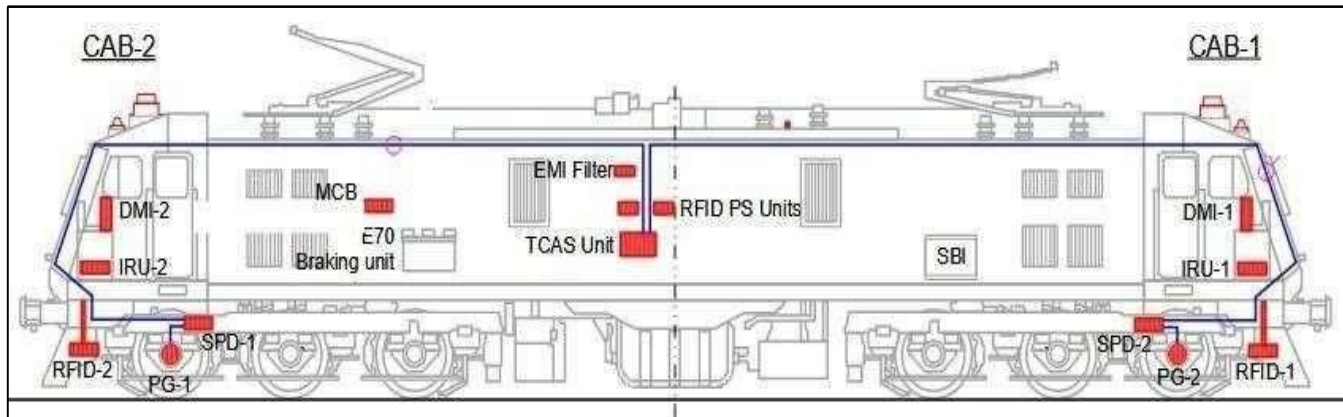


Figure 18: PG1&PG2 Assembly to Loco Axles

- 3.11.3 The screws heads shall be marked with green paint, after the QC inspector checks for proper torque application.
- 3.11.4 Ensure that tightness of the screws used for fixing of PG to loco axles and can be withstand for continuous running of locomotive.
- 3.11.5 Ensure that the PG head(sensor) is coming to top side from ground/rail or left/right side for safety purpose.
- 3.11.6 Ensure that PG cables routing made properly without hanging and tied properly with cable ties/Metal ties.
- 3.11.7 Ensure that the Speedo meter boxes are fixed nearby PG's.
- 3.11.8 Ensure that speedometer supporting clamps are welded without any gaps and cracks.



Figure 19: Speedometer Assembly

- 3.11.9 Ensure that speedometer fixing holes are matched with fixing clamps holes, uneven fixing may cause loose connection of speedo meter will leads to peel out from locomotive.
- 3.11.10 Ensure that cables from PG to Speedometers and from speedometer units to Loco Kavach unit are proper.
- 3.11.11 Ensure that PG1 connected to MC8 and PG2 connected to MC22 at loco Kavach unit.
- 3.11.12 Ensure that external cables are routed to their respective speedometer units.
- 3.11.13 Ensure that the external cable circular connectors are properly locked with speedometer box unit receptacles.



Figure 20: PG's & Speedometers Assembly

3.12 RFID Reader Assembly:

- 3.12.1 Check the RFID Reader brackets are welded bottom of the locomotive at both the CAB's.
- 3.12.2 Ensure that bracket's welding done without any gaps and joint breaks.
- 3.12.3 Ensure that welding done properly, such that the RFID Reader can be withstand for loco vibrations during running.
- 3.12.4 Check the tightness for channels fixing screws(M8X16mm) by using torque wrench by maintaining torque 25N-m to confirm that the tightness is good enough.
- 3.12.5 The installation of RFID reader would be done with the vertical distance from bottom of RFID reader to top of the rail level is $450\text{mm} \pm 50\text{mm}$, Maintain RFID reader installed at a distance of 1 to 3 meters from the end of cattle guard.
- 3.12.6 Ensure that all welded portions to be painted properly with pebble Grey color paint to avoid corrosion.

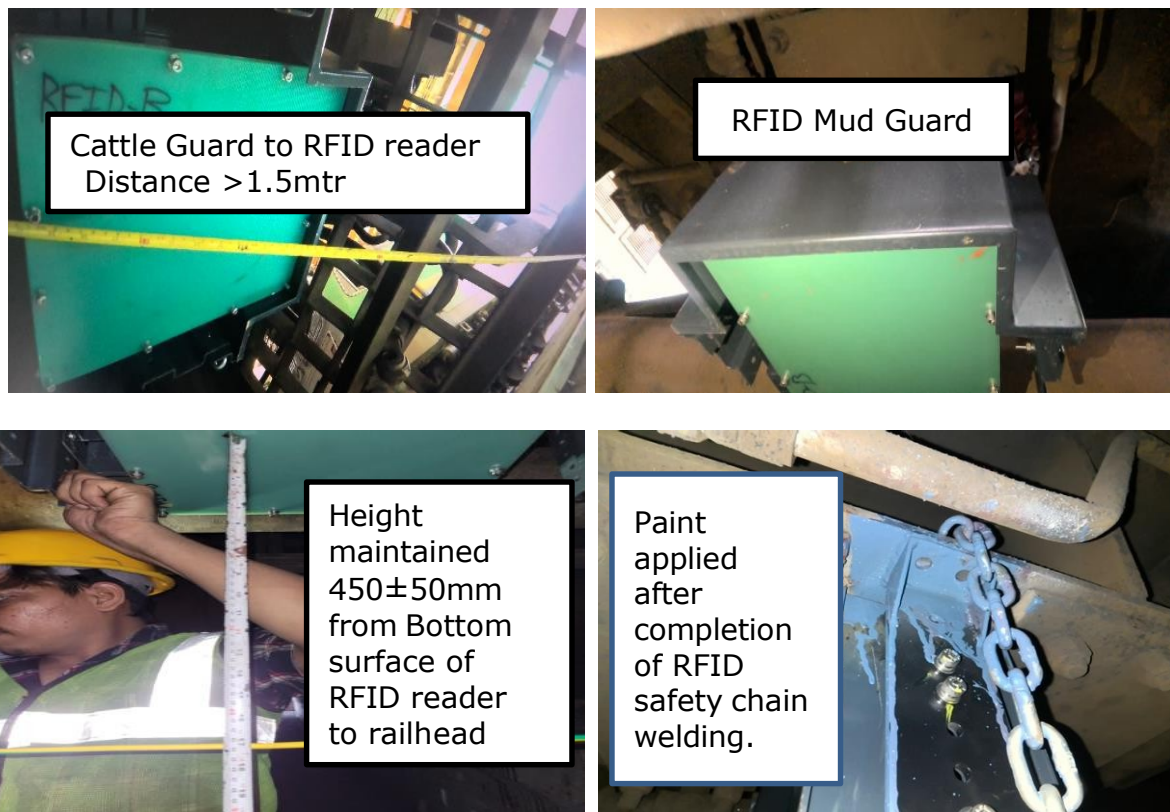


Figure 21: RFID Reader assembly

- 3.12.7 Check and Ensure that RFID reader cables are routed properly through the loco trench without any overlaps and over stress.
- 3.12.8 Is mud guard (from the Loco Kavach Installation Kit) been fixed to RFID reader mounting bracket, installed in front of each RFID reader on the cattle guard side, using the supplied bolts and washers.
- 3.12.9 Ensure that cable shall not have any sharp bends in routing and also ensure that cables shall not have any damages while routing.
- 3.12.10 Ensure that mil connector booting was not damage in routing.

- 3.12.11 Ensure that mill connectors connectivity is as per the wiring at loco side and RFID Reader side.
- 3.12.12 Ensure that RFID Reader-1 is connected to MC6 and RFID Reader-2 connected to MC7 at Loco Kavach.
- 3.12.13 Ensure that the circular connectors are properly locked with RFID box unit receptacles.



Figure 22: RFID Reader cable assembly and routing

3.13 Earthing:

- 3.13.1 Ensure that earthing done with 10Sq.mm Yellow/Green cable for Loco Kavach main unit, RIB unit, CAB Input Box, LP-OCIP units on both the cabs.
- 3.13.2 Ensure that cable continuity, lugs crimping and tightness of earth cable.
- 3.13.3 Ensure that the earth cables are routed through the conduit and routed properly and tied with cable ties.



Kavach Unit Earthing



RIB Earthing



CAB I/P Earthing

**Frame Earthing****Unit Earthing****DMI Earthing**

Figure 23: Earth cable connectivity for all units

3.14 Installation of CCB modules:

3.14.1 PSJB & TPM modules Installation:

- 3.14.2 On Loco CCB Brake panel, there is already installed Junction box, PSJB box and MPIO modules.
- 3.14.3 The factory supplied PSJB box shall be installed on the CCB brake panel by removing the existing PSJB box and the TPM module shall be installed on the right side of the MPIO module as shown in the below figure.
- 3.14.4 Ensure the installation shall be made with proper hardware provided along with the modules.
- 3.14.5 Ensure the connection for modules given, as per the labels mention on the harness cables.
- 3.14.6 Ensure to hand over the removed PSJB modules to Loco Shed officials.
- 3.14.7 If Existing PSJB dated on before 2020/03; We have to handover railway team and Install new factory supplied PSJB Unit.
 - 3.14.7.1 The SIFA valve will be supplied along with the mounting frame, the frame shall be welded under A9-DBC (Driver Brake Controller) on CAB-1 side and fix the 1/2" elbow fitting at SIFA valve IN port location on the mounting frame.

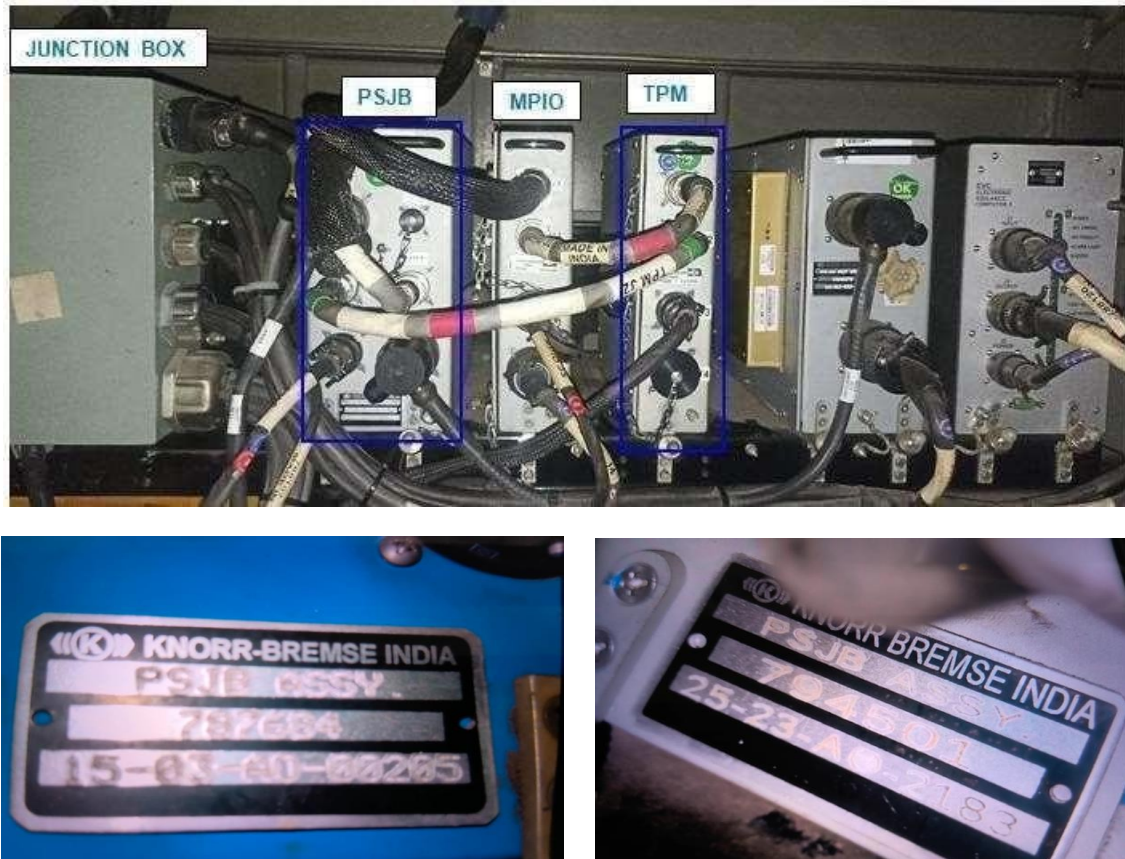


Figure 24: PSJB & TPM module Installation

- 3.14.7.2 Ensure the mounting location of the SIFA valve shall be easy for operations and maintenance.
- 3.14.7.3 Ensure the Isolation cock shall be fixed to SIFA valve for easy operation by loco pilots.
- 3.14.7.4 Ensure the copper pipes bending done with proper tools to avoid wrong bending or damages.
- 3.14.7.5 Ensure that SIFA VALVE SHOULD BE CLOSED CONDITION WHILE INSTALLATION.
- 3.14.7.6 Ensure the fixing of copper pipe using T-Joint(1/2") at BP pipe line and Elbow (1/2") at SIFA valve in port as shown in the below figure.





Figure 25: SIFA Valve Installation

3.15 Instructions for fastening with bolts and nuts:

- 3.15.1 All bolts and nuts shall be secured with suitable spring washers and plain washers. These washers are already supplied in Loco Kavach Installation kit, separately packed for each sub-system to be installed.
- 3.15.2 All threaded connections (bolts, nuts, screws etc.,) shall be secured with thread-locker, supplied in the Loco Kavach Installation kit.
- 3.15.3 Suitable torque spanners shall be used to apply torque appropriate to each screw, bolt or nut, per the table below and mark with (Yellow/Green) Paint.

Fastener size	Torque applied, N-m
M5	5
M6	10
M8	25
M10	40
M12	76

4. QC Report:

- 4.1.1 During the installation of Loco Kavach, the QC Engineer shall ensure the quality of installation and update the results in the QC in-process checklist (EG-IC-FT-34).
- 4.1.2 If any observations are found, they shall be recorded in the report and escalated to the Shed In-charge for necessary action. After completion of the rework, re-verification shall be carried out, and the status of rework along with the date shall be updated in the audit report.

In-Process QC Check List for Onboard Kavach Installation

QC Engineer Name: Rajkumar Bhaisare

v1.1

Loco ID	Loco Type	Brake Type	Railway Division	Shed Name	Inspection Date
38783	WAG-9HC	CCB	WCR	Itarsi(ETE)	18-09-2025

Total Points	Closed Points	Open Points
177	177	0

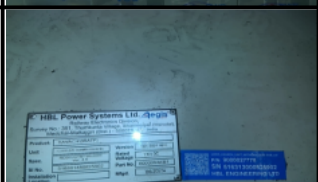
Start Date & Time	Completed Date & Time	Total Time
2025-09-17 09:52:32	2025-09-18 10:40:44	24h 48m 12sec

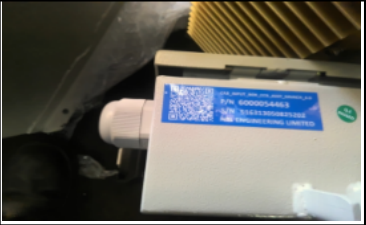
Completion Status	☒ Completed	☐ Not Completed
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1.0 DOCUMENT VERIFICATION

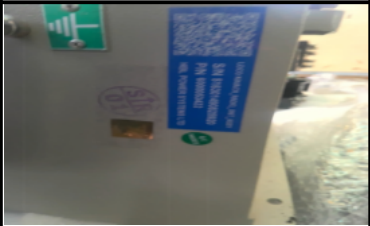
S.No	Description	Status	Remarks
1.1	Annexure of IC (Inspection Certificate) issued by RDSO.	Available	
1.2	Loco Allocation Letter	Available	

2.0 VERIFICATION OF SERIAL NUMBERS OF EQUIPMENT AS PER IC

S.No	Description	Status	Remarks	Image
2.1	Ensure presence of Hologram and S/R Stamp on each equipment	Present	Available on each equipment	
2.2	Loco KAVACH Main Unit - 516301480825902	Matching	Installed in CAB 1	

S.No	Description	Status	Remarks	Image
2.3	Relay Interface Box - 516313360825202	Matching	Installed in CAB 1	
2.4	Cab Input Box - 516313050825202	Matching	Installed in CAB 1	
2.5	RFID Reader 1 - 516300020825818	Matching	Installed in CAB 1	
2.6	RFID Reader 2 - 516300020825819	Matching	Installed in CAB 2	
2.7	LPOCIP (DMI) 1 - 516300010851769	Matching	Installed in CAB 1	
2.8	LPOCIP (DMI) 2 - 516300010851770	Matching	Installed in CAB 2	
2.9	Speedometer 1 - 516300030825937	Matching	Installed in near by wheel 5	
2.10	Speedometer 2 - 516300040825937	Matching	Installed in near by wheel 2	
2.11	GPS/GSM Antenna 1 - 516300330851918	Matching	Installed in CAB 1	

2.12	GPS/GSM Antenna 2 - 516300330851919	Matching	Installed in CAB 2	
2.13	UHF Radio Antenna 1 - 516311310825151	Matching	Installed in CAB 1	
2.14	UHF Radio Antenna 2 - 516311310825152	Matching	Installed in CAB 1	
2.15	UHF Radio Antenna 3 - 516311310825153	Matching	Installed in CAB 2	
2.16	UHF Radio Antenna 4 - 516311310825154	Matching	Installed in CAB 2	
2.17	RFID PS 1 - 516329200851763	Matching	Installed in CAB 1	
2.18	RFID PS 2 - 516329200851764	Matching	Installed in CAB 2	
2.19	Pulse Generator 1 - 516300350825057	Matching	Installed on wheel 5	

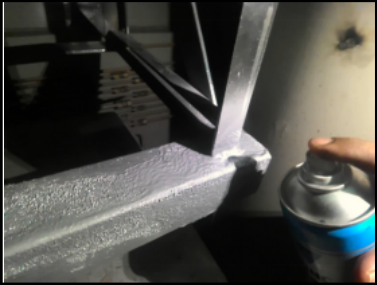
2.20	Pulse Generator 2 - 516300350825058	Matching	Installed on wheel 2	
2.21	PPC Card 1 - 516102100525709	Matching		
2.22	PPC Card 2 - 516102100525353	Matching		
2.23	VC Card 1 - 516101000551088	Matching		
2.24	VC Card 2 - 516101000551005	Matching		
2.25	VC Card 3 - 516101000525950	Matching		
2.26	Voter Card 1 - 516106100525794	Matching		
2.27	Voter Card 2 - 516106100525441	Matching		
2.28	Vital Gate Way Card 1 - 516174000525256	Matching		
2.29	Vital Gate Way Card 2 - 516174000525578	Matching		
2.30	Cab I/P Card 1 - 516110100525073	Matching		
2.31	Cab I/P Card 2 - 516110100525133	Matching		
2.32	DPS Card 1 - 516103410725179	Matching		
2.33	DPS Card 2 - 516103420725190	Matching		
2.34	Radio unit - 516301490825920	Matching	Installed in CAB 1	
2.35	EMI Filter Unit - 516300070825874	Matching	Installed in CAB 1	

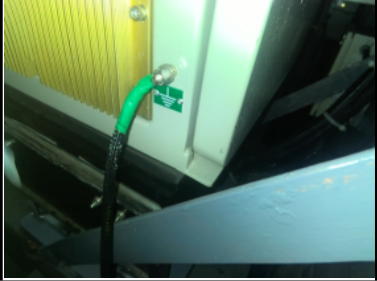
2.36	Radio Modem-1 - 203167	Matching	Installed in CAB 1	
2.37	Radio Modem-2 - 203295	Matching	Installed in CAB 1	
2.38	Interface Relay Unit Faiveley-1	Not Applicable		
2.39	Interface Relay Unit Faiveley-2	Not Applicable		
2.40	IRAB Main Unit	Not Applicable		

3.0 LOCO KAVACH

Start Date & Time	Completed Date & Time	Total Time
17/9/2025, 4:01:44 pm	18/9/2025, 10:34:57 am	18h 33m 13sec

S.No	Description	Status	Remarks	Image
3.1	Placement of LOCO KAVACH Equipment in Locomotive: Verify that all Loco KAVACH equipment and peripherals are installed and connected as per connectivity diagram	Connected	Verified that all Loco KAVACH equipment and peripherals are installed and connected	
3.2	Ensure that adequate space is available around the Loco KAVACH for ease of maintenance and service.	Available	Adequate space is available	
3.3	Loco Stand: Loco Kavach Unit has been placed on the designated stand and secured using the mounting bolts supplied in the Loco Kavach Installation Kit?	Yes		

3.4	Welding between Loco surface and KAVACH fixing Stand:Inspect the welding between the Loco surface and the KAVACH stand. Ensure there are no sharp edges or gaps. Ensure 2.80 mm diameter E6013 welding electrodes are used.	Yes	Welding between Loco surface and KAVACH fixing Stand has been completed and verified, welds are continuously properly aligned And Ferroseal 3.15mm diameter E6013 Electrodes are used in welding.	
3.5	Welding Surface Treatment: Is Aerol Zinc3060 sprayed and the coating is extended up to 50 mm on both sides?	Yes	Aerol Zinc3060 spray is applied	
3.5.1	Welding Surface Treatment: Verify whether second coat of Aerol Zinc 3060 is done over entire area? After 30 Minutes Gap	Yes	Second coat of Aerol Zinc 3060 is applied After 30 minutes gap	
3.5.2	Welding Surface Treatment: Verify whether Berger Red Oxide Primer is applied on all galvanized steel parts?	Yes	Berger Red Oxide Primer is applied on all galvanized steel parts	
3.6	Loco Stand fixing holes:Ensure that Loco stand fixing holes(M8) are exactly matching with Loco channel fixing holes(M8).	Matching	Loco stand fixing holes(M8) are exactly matched	
3.7	Torque and Marking:Verify the torque Value of M8 Bolts (25 N.M) and mark with green/Yellow paint if the torque value is Ok.	Torquing done	Torque Value of M8 Bolts (25 N.M) applied and marked with yellow paint	

3.8	Cable connections:Verify the connections on the Loco external cable are made correctly according to the "Loco KAVACH External Harness Connectivity Diagram" without any overlaps.	Connected		
3.9	Wire Stress:Routing of wires to be done without stress and without sharp bends.	Routing Done	Routing of wires to be done without stress and without sharp bends	
3.10	Cable securing:Verify that all peripheral cables for the loco kavach unit are routed and securely fastened using the appropriate metal clamps.	Metal clamps implemented		
3.11	Connector fitment:Ensure that all the external cable circular connectors are fully locked properly with Loco Kavach receptacles.	Cables Connected	All the external cable circular connectors are fully locked properly with Loco Kavach receptacles	
3.12	Earthing:Ensure that earthing done with 10Sq.mm Yellow/Green cable for Loco Kavach main unit.	Yes	Earthing done with 10Sq.mm Yellow/Green cable	
3.13	Earth Cable:Ensure that cable continuity,lug's crimping and tightness of earth cable.	Yes		

3.14	Earth Cable routing:Ensure that the earth cables are routed through the conduit and routed properly and tied with metal clamps / cable ties.	Routing Done	Earth cables are routed through the conduit and routed properly and tied with cable ties	
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4.0 EMI FILTER BOX

Start Date & Time	Completed Date & Time	Total Time
17/9/2025, 4:03:24 pm	17/9/2025, 10:57:09 pm	6h 53m 45sec

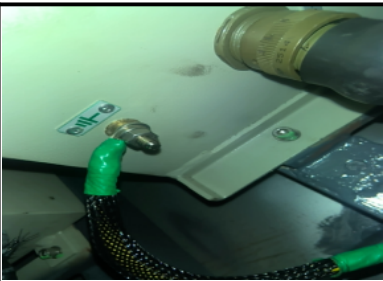
S.No	Description	Status	Remarks	Image
4.1	EMI Filter Box Fixing:Fix M5x16mm Bolts on the EMI Filter Box to the loco stand with torque of 6 N-m and mark with green/yellow paint.	Fixed	Used M5 bolt torque applied 6N.M;marked with yellow paint	
4.2	Cable routing:Ensure all the cables routed through PG gland without sharp bends, stress and tied with metal clamps.	Routing Done	Cables routed through PG gland without sharp bends, stress and tied with metal clamps	

5.0 RIB AND CAB INPUT BOX

Start Date & Time	Completed Date & Time	Total Time
17/9/2025, 4:03:52 pm	17/9/2025, 10:57:34 pm	6h 53m 42sec

S.No	Description	Status	Remarks	Image
5.1	Space between RIB and CAB Input:Ensure enough space available in-between cab input box and RIB unit for easy access of cables	Yes	Enough space available in-between cab input box and RIB unit	

5.2	Welding:Ensure RIB and CAB Input box stand welding is without gap, cracks and joint breaks.	Yes	Welding observed properly; No gaps & cracks	
5.3	Torque:The M5X16mm bolts shall be tightened with 6 N.m torque Wrench and mark with green/yellow paint after torque verification.	Yes	Used M5 bolt torque applied 6N.M;marked with Yellow paint	
5.4	Cable Connections:Verify the RIB Unit harness cable connections for MILB1, MILB2, MILB3, MC26, MC18 are properly terminated.	Yes	RIB Unit harness cable connections for MILB1, MILB2, MILB3, MC26, MC18 are properly terminated	
5.5	Cable Routing:Verify proper routing through PG glands without sharp bends or stress. Cables must be tied using metal clamps.	Routing Done		
5.6	Connector fitment:Verify that cables are connected with respect to labels and ensure all the external cable circular connectors are fully locked properly with enclosure (LOCO KAVACH) receptacles.	Connected		
5.7	Earthing:Ensure that earthing done with 10Sq.mm Yellow/Green cable for Relay Interface Box and Cab Input box.	Yes	Earthing done with 10Sq.mm Yellow/Green cable	

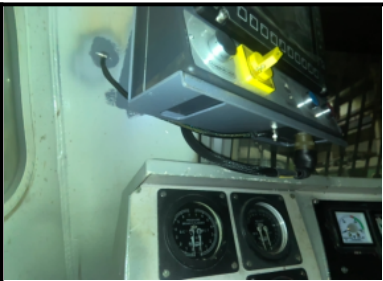
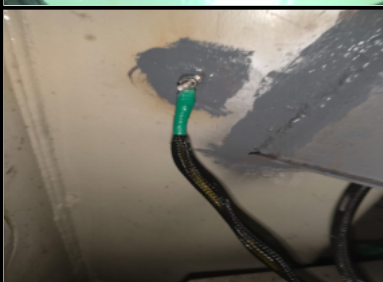
5.8	Earth Cable:Ensure that cable continuity, lug's crimping and tightness of earth cable.	Yes		
5.9	Earth Cable routing:Ensure that the earth cables are routed through the conduit and routed properly and tied with metal clamps / cable ties.	Routing Done	Earth cables are routed through the routed properly and tied with cable ties.	

6.0 DMI (LP-OCIP)

Start Date & Time	Completed Date & Time	Total Time
17/9/2025, 4:01:26 pm	17/9/2025, 11:02:08 pm	7h 0m 42sec

S.No	Description	Status	Remarks	Image
6.1	DMI Mounting Place:Check the DMI mounted place is good enough at driver desk, which can be operated easily by Loco pilot.	Available	DMI mounted place is good enough at driver desk	
6.2	DMI Mounting Stand:Ensure that DMI mounting stand is properly welded without any joint gaps and can be withstand to loco vibrations.	Yes	DMI mounting stand is properly welded without any joint gaps	
6.3	Torque and Marking:Ensure M5x16mm Bolts are tightened to 6 N·m torque using a torque wrench, and mark screw heads with green/yellow paint after torque verification.	Torquing done	Used M5 bolt torque applied 6N.M;marked with Yellow paint	

6.4	DMI Cable:Make sure the DMI cable can be easily accessed by the projection at the bottom of the stand.	Yes		
6.5	DMI Stand Color:Ensure welded portions of the stand are treated with red oxide coating before applying RAL7032 (Pebble Grey/ Smoke Gray) paint to prevent corrosion	Applied	Smoke Grey paint applied	
6.6	Cable routing:Ensure all the cables routed without any stress, without sharp bends and tied with metal clamps.	Routing Done	All the cables routed without any stress, without sharp bends	
6.7	Cable Booting:Verify that the cable booting is not damaged or peeled during and after cable routing, and ensure circular connectors are fully locked with the DMI unit enclosure receptacles	Yes		
6.8	DMI-1 Cable Connection:DMI-1 cable should be connected to MC1 at Loco Kavach unit.	Connected	DMI-1 cable connected to MC1	
6.9	DMI 2 Cable Connection:DMI-2 cable should be connected to MC3 at Loco Kavach unit.	Connected	DMI-2 cable connected to MC3	

6.10	Earthing:Ensure that earthing done with 10Sq.mm Yellow/Green cable for LP-OCIP-1 and LP-OCIP-2.	Yes	Earthing done with 10Sq.mm Yellow/Green cable	
6.11	Earth Cable:Ensure that cable continuity,lug's crimping and tightness of earth cable.	Yes		
6.12	Earth Cable routing:Ensure that the earth cables are routed through the conduit and routed properly and tied with metal clamps / cable ties.	Routing Done	Earth cables are routed through the routed properly and tied with cable ties	

7.0 RFID PS UNIT

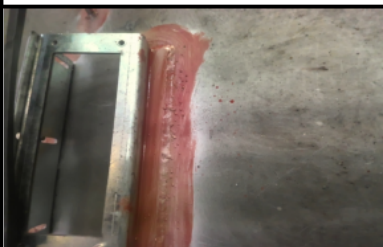
Start Date & Time	Completed Date & Time	Total Time
17/9/2025, 4:20:59 pm	17/9/2025, 10:58:10 pm	6h 37m 11sec

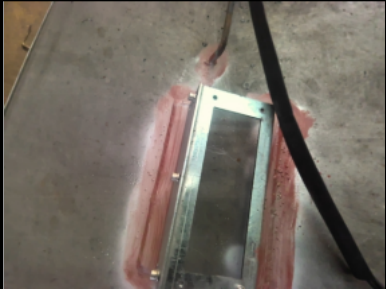
S.No	Description	Status	Remarks	Image
7.1	RFID PS Unit fixing:Ensure M5x16mm Bolts are tightened to 6 N·m torque using a torque wrench on the Loco Kavach stand, and mark screw heads with green/yellow paint after verifying the torque value	Torquing done	Used M5x16mm Bolts are tightened to 6 N·m, marked with Yellow paint	
7.2	Cable connections:Ensure that cable connections are given at the corresponding location with respect to labels provided at Loco Kavach.	Yes		

7.3	Cable routing:Ensure that cables are routed properly without hanging and tied properly with cables ties	Routing Done	Cables are routed properly without hanging and tied properly with cables ties	
7.4	Connector locking:Ensure the circular connectors are properly locked with RFID box unit receptacles.	Locked	Connectors are properly locked with RFID box unit receptacles	

8.0 LOCO ANTENNA AND GPS/GSM ANTENNA

Start Date & Time	Completed Date & Time	Total Time
17/9/2025, 4:10:28 pm	17/9/2025, 10:58:34 pm	6h 48m 6sec

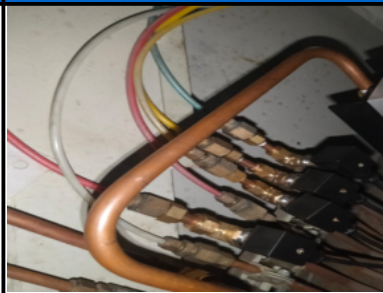
S.No	Description	Status	Remarks	Image
8.1	Are RF and GPS/GSM antennas properly leveled using a spirit level instrument?	Yes	GPS/GSM antennas properly leveled using a spirit level	
8.2	Radio Antenna Welding:Check that the radio antenna's base plate welding is done properly, with no joint gaps or cracks.	Yes	Radio antenna's base plate welding is done properly, with no joint gaps or cracks	

8.3	Welding:Ensure the welding is sufficient to withstand locomotive vibrations at higher speeds without affecting the antennas.	Yes		
8.4	Antenna Mounting:Antennas are to be mounted within stipulated height to avoid OHE line contact.	Yes		
8.5	RF Antenna Cable Connections:Are the Rx and Tx cables connected to their respective Rx and Tx antenna, as per the labels?	Connected	Rx and Tx cables connected to their respective Rx and Tx antenna	
8.6	GPS/GSM Antenna and Cable Installation:Are the GPS and GSM cables clearly labeled and correctly connected to their respective GPS and GSM antenna on both sides of the locomotive?	Connected	GPS and GSM cables clearly labeled and correctly connected to their respective GPS and GSM antenna on both sides of the locomotive	
8.7	Antenna Cables Routing: Verify that cables from the two RF antenna and GPS/GSM antennae are routed through a 2" steel-reinforced conduit.	Routing Done	GPS/GSM are routed through a hole under the antenna	
8.7.1	Antenna Cables Routing: Ensure the conduit is securely clamped to the roof using clamps welded to the rooftop, as per the installation drawing.	Yes		
8.7.2	Antenna Cables Routing: Confirm the conduit is routed into the Loco cabin through the elbow pipe installed in the flasher unit on the roof.	Yes		
8.7.3	Antenna Cables Routing: Ensure the conduit pipe and elbow are sourced from the Loco Kavach Installation Kit.	Yes		

8.7.4	Antenna Cables Routing: Verify that RTV Silicone compound (from the Loco Kavach Installation Kit) is applied around all open conduit and cable entry joints to prevent ingress of dust,water, etc.	Yes	RTV Silicone compound (from the Loco Kavach Installation Kit) is applied	
8.8	Red oxide coating:Ensure that all welded portions should be treated with red oxide coating, before painted with RAL7032 (pebble grey / Smoke Gray) paint, to avoid corrosion.	Applied	Smoke Grey applied	

9.0 PNEUMATIC FITTINGS AND EP VALVE COCKS

Start Date & Time	Completed Date & Time	Total Time
17/9/2025, 4:21:24 pm	17/9/2025, 10:58:51 pm	6h 37m 27sec

S.No	Description	Status	Remarks	Image
9.1	Pneumatic fittings:Confirm that all pipes and fittings used in the assembly are from the approved BOM and sourced from the I and C kit supplied by the factory.Any locally procured items must be checked for compliance with the approved BOM	Yes	All pipes and fittings used in the assembly are from the approved BOM and sourced from the I and C kit supplied by the factory	
9.2	Copper Pipes:Ensure that copper pipes are bent using appropriate bending tool, and there no kinks or sharp bends in the pipe.	Yes	Copper pipes are bent using hook alignment mark tool	
9.3	Copper Tube:Ensure that copper tube length is measured with respect to the connectivity from loco pneumatic to EP Valve, BP cock, Horn cock and valve arrangements.	Yes		

9.4	Copper pipe connections:Ensure that copper pipe connections made properly with approved make (Ex. Fluid Control) ferrules and TEE-joints used.	Yes	Copper pipes are connected properly and TEE-joints used	
9.5	Threaded Connections:All threaded connections must be sealed with loctite 567 (Not With Teflon Tape), which is supplied in the landC kit.	Yes	Loctite 567 applied	
9.6	Pneumatic Lines Connections:Check the pneumatic lines with a soap solution to make sure there are no loose connections and no air bubbles should be seen.	Yes	Soap solution test observed no loose connections and no air bubbles	
9.7	Pneumatic Lines Connections:The soap solution must be cleaned after the test.	Yes	Soap solution cleaned	

10.0 PRESSURE SENSORS

Start Date & Time	Completed Date & Time	Total Time
17/9/2025, 5:25:18 pm	17/9/2025, 10:59:01 pm	5h 33m 43sec

S.No	Description	Status	Remarks	Image
10.1	Ensure all these pressure sensors shall be installed under CAB1 driver desk.	Installed	Pressure sensors are installed under CAB1	
10.2	Ensure MR sensor should be 16 bar, and remain BP,BC1,BC2 are 7 bar	Installed	MR sensor 16 bar, and remain BP,BC1,BC2 are 7 bar	

10.3	Ensure all pressure sensors should be installed on T-Joints	Installed	Pressure sensors are installed on T-Joints	
10.4	Wiring Connections : Red (+V) to Terminal 1, Black (–V) to Terminal 2 & Black-White (Earth) to Terminal 3—Mention In Status	Connected	Red (+V) to Terminal 1, Black (–V) to Terminal 2 & Black-White (Earth) to Terminal 3 are connected	

11.0 IRU FAVIELY UNITS FIXING FOR E70 TYPE LOCO

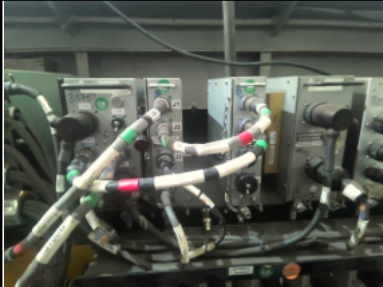
Start Date & Time	Completed Date & Time	Total Time
17/9/2025, 4:44:31 pm	17/9/2025, 4:44:38 pm	0h 0m 7sec

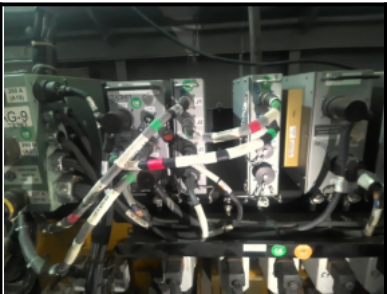
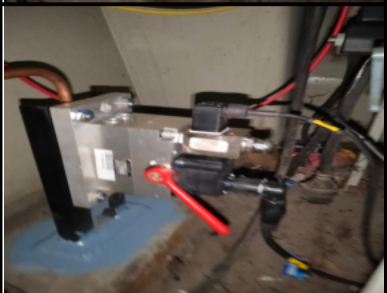
S.No	Description	Status	Remarks	Image
11.1	IRU Fixing Place:Are two IRUs installed—one in CAB-I and one in CAB-II—as per installation instructions?	Not Applicable		
11.1.1	IRU Fixing Place: Is the mounting frame fabricated using 50x50x5 mm galvanized steel tube or angle?	Not Applicable		
11.2	Welding:Has the mounting frame been welded under the A9 Driver Brake Controller (DBC) of each cab and after welding has the frame been painted properly?	Not Applicable		
11.3	Red oxide coating:Ensure that the welded portion should be coated with red oxide and painted with RAL7032 (Pebble grey/smoke gray) paint.	Not Applicable		
11.4	Have the IRU units been fixed to the mounting frame using the bolts, nuts, and washers supplied in the Brake Interface Installation Kit?	Not Applicable		
11.5	Cable connections:Ensure that cable connections are given at the corresponding location with respect to labels provided.	Not Applicable		
11.6	Cable routing:Ensure that cables are routed properly without hanging on ground and tied properly with cable ties.	Not Applicable		
11.7	EP Valve and Isolation Cock are installed on the stand where the IRU unit is mounted -- Mention In Status	Not Applicable		
11.8	EP Solenoid Valve wiring: Red (+V) connected to Terminal 1, Black (–V) connected to Terminal 2 (Cable from MILB3)-- Mention In Status	Not Applicable		

11.9	Isolation Cock (N/C type) wiring: Red (+V) connected to Terminal 1, Black (–V) to Terminal 2 (Cable from MC26)-- Mention In Status	Not Applicable		
11.10	Verify that the existing D2 Relay Valve and Manifold Unit are removed from previous location and New D2 Relay Valve is installed on the LE (Locomotive Equipment) unit ----- Mention In Status	Not Applicable		
11.11	Wire connections: LE Unit Solenoid Valve wiring: Red (+V) connected to Terminal 1, Black (–V) to Terminal 2 (Cable from MILB1)-- Mention In Status	Not Applicable		

12.0 PSJB AND TPM UNITS FIXING FOR CCB TYPE LOCO

Start Date & Time	Completed Date & Time	Total Time
17/9/2025, 4:23:03 pm	18/9/2025, 10:35:25 am	18h 12m 22sec

S.No	Description	Status	Remarks	Image
12.1	PSJB Fixing:Ensure existing PSJB removed and Handed over to Loco Shed Rail team, And install factory supplied PSJB	Installed	From railway side already modified PSJB was installed.So we did not installed factory received PSJB.	
12.2	TPM Unit Fixing:Ensure that TPM module is installed right side to the MPIO module	Installed	The TPM module is installed on the right side of the MPRO module the MPIO module	
12.3	Fixing of PSJB and TPM:Ensure that PSJB and TPM units fixing holes are matching with supporting clamps	Matching	The PSJB and TPM units fixing holes are matching with supporting clamps	

12.4	Cable connections:Ensure that cable connections are given at the corresponding location with respect to labels provided.	Connected	Endure the cable are connected to corresponding location with respect to labels provided	
12.5	Cable routing:Ensure that cables are routed properly without hanging on ground and tied properly with cable ties.	Routing Done	cables are routed properly without hanging on ground and tied properly with cable ties.	
12.6	Ensure SIFA Valve is installed with mounting frame under DBC panel in CAB-1 side(NOTE: Ensure that SIFA VALVE SHOULD BE CLOSED CONDITION WHILE INSTALLATION)	Installed	SIFA Valve is installed with mounting frame under DBC panel in CAB-1 side	
12.7	Ensure that mounting location of the SIFA valve should be easy for operations and maintenance	Yes	Ensure the mounting location of the SIFA valve should be easy for operations and maintenance	
12.8	Ensure SIFA valve manifold should be fixed to mounting frame by using hardware provided along with the installation Kit.	Fixed	The SIFA valve manifold should be fixed to mounting frame by using hardware provided along with the installation Kit.	
12.9	Welding: Ensure that welded portion should be neat and clean. Check that there is no welding gaps and cracks.	Yes	Welded portion properly welding with nest and clean & checked with welding there is no welding gaps and cracks	

13.0 IRAB (Indian Railway Air-Break)

Start Date & Time	Completed Date & Time	Total Time
17/9/2025, 4:44:46 pm	17/9/2025, 4:44:54 pm	0h 0m 8sec

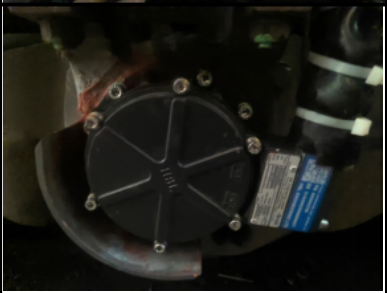
S.No	Description	Status	Remarks	Image
13.1	Check the Pneumatic connections are as per the connectivity diagram.	Not Applicable		
13.2	Verify ball valve installation on BP(3/4") and MR (3/8") pipes, and confirm Loctite 567 is used at every connection.	Not Applicable		
13.3	Ensure BC pressure transducer reads 7 bar, BP is at 5 bar, and MR is at 16 bar.	Not Applicable		
13.4	Check that all pipe joints are properly connected to the LPSR (MUB-2 and MUB-3).	Not Applicable		
13.5	Check that the BP line is connected to the QRV (Quick Release Valve).	Not Applicable		
13.6	Check if the ball valve electrical terminations are properly connected.	Not Applicable		

14.0 PG AND SPEEDO METER UNITS

Start Date & Time	Completed Date & Time	Total Time
17/9/2025, 10:16:59 am	17/9/2025, 11:00:19 pm	12h 43m 20sec

S.No	Description	Status	Remarks	Image
14.1	PG1 and PG2 Installation:Ensure that PG1(Left from LP of CAB1) and PG2 (Right from LP of CAB1) are installed on allotted axles (WAP5=wheel 2and3, WAP7=wheel2and5) of locomotives on left and right side.	Installed	Installed PG -1 & PG - 2	
14.2	Washer Insertion:Verify that M12 spring is correctly inserted in the drive pin.	Yes	M12 spring is correctly inserted in the drive pin	

14.3	Drive Pin Fixing: Ensure drive pin length before installation as per Loco type, (loco type WAP7=60 mm, WAP5=76mm, EMU=90mm). Ensure the drive pin is securely fixed with LOCTITE 542. Verify that the torque applied is 76N·m using a calibrated torque wrench.	Torquing done	Drive pin installed, torque applied 76N.M	
14.4	Gasket Lock Plate Placement: Ensure the Gasket Lock Plate is correctly positioned on the axle cover without any misalignment.	Positioning done	Gasket Lock Plate is correctly positioned on the axle cover	
14.5	PG Coupler Ring Placement: Verify that PG coupler ring is positioned correctly on the gasket. Ensure that the label (PG1 / PG2) is clearly visible at the top end.	Positioning done	PG coupler ring is positioned correctly on the gasket	
14.6	Coupler Ring Alignment: Confirm that the selected PG (PG1 / PG2) determines the alignment of four out of eight holes with the axle cover.	Yes	PG (PG1 / PG2) determines the alignment of four out of eight holes with the axle cover is fitted	
14.7	Coupler Ring Fixing: Ensure the CSK Hex Socket screws size M10×30mm-SS are used to fix the coupler ring with LOCTITE 542. Check that screws are tightened to the specified torque of 40 N.m using a calibrated torque wrench. (Note: Bolthead surface should be flush.)	Torquing done	CSK Hex Socket screws size M10×30mm-SS are used to fix the coupler ring with LOCTITE 542 applied, torque of 40 N.m	
14.8	Gasket PG Placement: Verify that "Gasket PG" is correctly placed on the coupler ring without any gaps or misalignment.	Yes		

14.9	PG1 / PG2 Insertion: Ensure PG1 / PG2 is inserted correctly into the axle cover. Confirm that driving fork is aligned with the drive pin before proceeding.	Aligned		
14.10	PG Alignment: Ensure that all eight holes on PG align perfectly with holes on the gasket and coupler ring.	Aligned	All eight holes on PG align perfectly with holes on the gasket and coupler ring	
14.11	PG Fixing with M8 Bolts: Check that the M8×25mm bolts are properly installed with LOCTITE 542. Ensure both spring and plain washers are used.	Installed	Used M8 bolts are properly installed	
14.12	M8 Bolt Torque Check: Verify that all M8 bolts are tightened to 25 N.m using a calibrated torque wrench. Marking with green/yellow paint.	Torquing done	M8 bolts are tightened to 25 N.m using a calibrated torque wrench, Marking with yellow paint	
14.13	Final Position Check: Ensure that the final installed positions of PG1 and PG2 match the specified locations.	Matching		
14.14	Cable routing: Ensure that PG cables routing made properly with metal clamps / cable ties	Routing Done	PG cables routing made properly with cable ties	

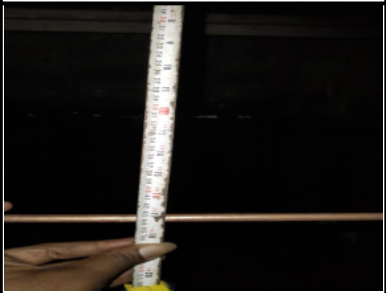
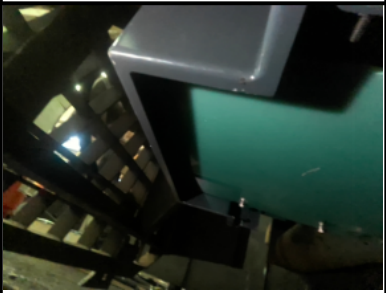
14.15	Speedometer Boxes fixing: Is each Speedometer Interface Unit mounted on the locomotive chassis near its corresponding Pulse Generator?	Connected		
14.15.1	Speedometer Boxes fixing: Is the Speedometer Interface Unit orientation such that cables connect to the Pulse Generator on one side and the Loco Kavach Unit on the other side without crisscrossing?	Connected		
14.15.2	Speedometer Boxes fixing: Speedometer holes and fixing clamp holes are to be matched evenly. Verify the M6 bolts torque 10 N-M. Marking with green/Yellow paint	Torquing done	Verified the M6 bolts torque applied 10 N-M. Marking with Yellow paint is applied	
14.16	Welding of supporting clamps: Ensure that speedometer supporting clamps are welded without any gaps and cracks.	Welding done		
14.17	Speedometer Cables: Ensure that cables from PG to Speedometers and from speedometer units to Loco Kavach unit are connected properly.	Connected		
14.18	Pulse Generator & Speedometer Interface Unit Connection: Is PG1 connected to the Speedometer Interface Unit with Part No.6000052976?	Connected	PG1 connected to the Speedometer Interface Unit with Part No.6000052976	

14.18.1	Pulse Generator and Speedometer Interface Unit Connection: Is PG2 connected to the Speedometer Interface Unit with Part No.6000052977?	Connected	PG2 connected to the Speedometer Interface Unit with Part No.6000052977	
14.19	Is each Speedometer Interface Unit correctly connected to the Loco Kavach Unit?	Connected		
14.20	Cable routing:Ensure that external cables are routed to their respective speedometer units	Routing Done		
14.21	Connector locking :Ensure that the external cable circular connectors are properly locked with speedometer box unit receptacles.	Locked	External cable circular connectors are properly locked with speedometer box unit.	

15.0 RFID READER ASSEMBLY

Start Date & Time	Completed Date & Time	Total Time
17/9/2025, 4:29:25 pm	18/9/2025, 10:19:04 am	17h 49m 39sec

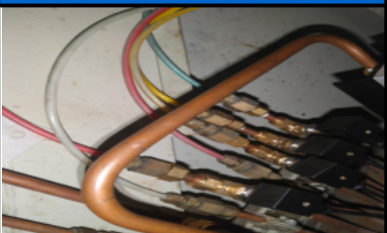
S.No	Description	Status	Remarks	Image
15.1	Is each RFID reader installed at a distance of 1 to 3 meters from the end of cattle guard?	Installed	RFID reader installed at a distance of 1.1meters from the end of cattle guard	

15.2	Welding quality:Ensure Stud welding/Arc Welding is done properly, such that the RFID Reader can be withstand for loco vibrations during running and Carried out DPT Test.	Yes	Arc Welding is done properly	
15.2.1	Welding quality: Has Ballata been placed between the RFID bracket and Loco chassis frame,M12 bolts tightened with Nylock nuts,torque verified at 20 N-m, and cotter pin inserted into the stud hole?	Not Applicable		
15.3	Channels tightness:Verify the tightness for channels fixing screws (M8X16mm) by using torque wrench (25N-M).marking with green/yellow pain	Torquing done	Used M8 bolt torque applied 25N.M, Marked with yellow paint	
15.4	Has the bottom surface of the RFID reader been adjusted and fixed at a height of 450 ± 50 mm from the rail head?	Yes	RFID reader been adjusted and fixed at a height of 420mm from the rail head	
15.5	Is one end of the Chain/Sling welded to locomotive chassis, away from the mounting bracket weld joint?	Yes	One end of the Chain is welded to locomotive chassis	
15.5.1	Is other end of the Chain/Sling securely fastened to RFID reader by using bolt provided in the Loco Kavach Installation Kit?	Yes	Other end of the Chain is securely fastened to RFID reader by using bolt	
15.6	Is mud guard (from the Loco Kavach Installation Kit) been fixed to RFID reader mounting bracket,installed in front of each RFID reader on the cattle guard side, using the supplied bolts and washers?	Fixed	Mud guard installed in front of each RFID reader on the cattle guard side, using the supplied bolts and washer	

15.7	Cable trench:Verify that RFID reader cables are routed properly through the loco trench without any overlaps and over-stress. Ensure that cable should not have any sharp bends in routing and mill connectors without any damage while routing.	Routing Done		
15.8	Connector connectivity:Ensure that MIL connectors connectivity as per the connectivity drawing.	Connected		
15.9	RFID -1 connectivity:Ensure that RFID Reader-1 is connected to MC6 at Loco Kavach.	Yes	RFID Reader-1 is connected to MC6 at Loco Kavach.	
15.10	RFID -2 connectivity:Ensure that RFID Reader-2 is connected to MC7 at Loco Kavach.	Yes	RFID Reader-2 is connected to MC7 at Loco Kavach.	
15.11	Circular connectors:Ensure that the circular connectors are properly locked with RFID box unit receptacles.	Locked	Connectors are properly locked with RFID box unit	

16.0 AUTO HORN INSTALLATION

Start Date & Time	Completed Date & Time	Total Time
17/9/2025, 4:21:41 pm	17/9/2025, 11:00:58 pm	6h 39m 17sec

S.No	Description	Status	Remarks	Image
16.1	Pneumatic fittings:Confirm that all pipes and fittings used in the assembly are from the approved BOM and sourced from the landC kit supplied by the factory.	Yes	All pipes and fittings used in the assembly are from the approved BOM and sourced from the I&C kit supplied by the factory	

16.2	Copper Pipes: Confirm that copper pipes are bent using appropriate bending tool, and that there no kinks or sharp bends in the pipe.	Yes	Copper pipes are bent using hook alignment mark tool	
16.3	Copper Tube :Ensure that copper tube length is measured with respect to the connectivity from loco pneumatics MR to ON/OFF Ball Cock, Solenoid Valve to HT Horn MR Pipe arrangements, as per approved drawing for WAP5, WAP7, WAG9, WAG7 and WAP4. This pneumatic arrangement is not required for Diesele locos.	Yes		
16.4	Copper pipe connections:Ensure that copper pipe connections made properly with approved make (Ex. Fluid Control) ferrules and TEE-joints used.	Yes	Copper pipes are connected properly and TEE-joints used	
16.5	Auto Horn Solenoid valve connections(Red +ve) to be connected at terminal1 and (Black -ve) to be connected at terminal2. This cable part of CAB I/P wiring of TB21 (Red +ve) & TB22 (Black -ve). This cable part of CAB I/P wiring of TB21 (Red +ve) & TB22 (Black -ve).	Connected		

Note : This QC report is generated by the system and does not need a physical signature.

Signature: Prabhakar Gundu

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Pradeep Kongalla (Sep 30, 2025 13:56:00 GMT+5.5)

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InProcess QC procedure for On Board KAVACH Locomotives_V1.4(New)

Final Audit Report

2025-09-30

Created:	2025-09-30 (India Standard Time)
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"InProcess QC procedure for On Board KAVACH Locomotives_V1.4(New)" History

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